

Lab Class NLP:VIII

Exercise 1 : Word2Vec

Word2vec is widely used to produce word representations, but it has some shortcomings.

- (a) Word2vec suffers from out-of-vocabulary problems. Explain a technique that you might use to mitigate them.
- (b) Suppose that you are tasked with building a job recommendation system. You could use pre-trained word embeddings to match applicants to job postings based on a personal profile. What ethical problems might arise if you did this?
- (c) Explain a technique that you might use to mitigate these problems.

Exercise 2 : Skip-Gram Model

The skip-gram model learns embeddings for individual words. In this model we are given a corpus of words w and their contexts c . We consider the conditional probabilities $p(c|w)$, and our goal is to set the parameters θ of $p(c|w; \theta)$ so as to maximize the probability:

$$\operatorname{argmax}_{\theta} = \prod_{(w,c) \in D} p(c|w; \theta)$$

where D is the set of all word and context pairs we extract from the text.

- (a) Write down how $p(c|w; \theta)$ is estimated using a softmax classifier. What are the parameters of this model?
- (b) Why is it computationally expensive to compute the softmax objective? Give a high level description of why the negative sampling approach is a more efficient way of deriving word embeddings.
- (c) Skip-gram learns representations for individual words. In many NLP applications however, we work with sentences, paragraphs, or even entire documents. How could you modify skip-gram in order to learn representations for entire documents in addition to embeddings for context and target words? Draw a picture of your modified skip-gram model. Now write down the negative sampling objective with your new skip-gram model.